Count equal groups in the sunshine

Shadows at Noon

Name _____ Date _____

Show your thinking with numbers, marks, or words.

1. Little draws 2 rows of chalk marks on the path, with 3 marks in each row. How many chalk marks are there?

$3 + 3 =$ _____

Show your thinking.

2. Big sets out 3 groups of measuring pebbles, with 4 pebbles in each group. How many pebbles are there altogether?

$4 + 4 + 4 =$ _____

Show your thinking.

3. Little lays sticks in 4 rows in the sunny yard, with 2 sticks in each row. How many sticks are there?

$2 + 2 + 2 + 2 =$ _____

Show your thinking.

Count equal groups in the sunshine

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4. A shadow notebook has 2 rows of boxes, with 5 boxes in each row. How many boxes are there?

$5 + 5 =$ _____

Show your thinking.

5. Little draws 5 groups of shadow-tip dots in the notebook, with 3 dots in each group. How many dots are there?

$3 + 3 + 3 + 3 + 3 =$ _____

Show your thinking.

6. Big draws 5 rows of chalk circles beside the shadow, with 5 circles in each row. How many circles are there?

$5 + 5 + 5 + 5 + 5 =$ _____

Show your thinking.

WONDER DAILY

Shadows at Noon

The shadow that changed



A short shadow. A high sun.

GRADE 3

Little Wonderer found a long shape beside a standing stick. It was a shadow: a place where the stick blocked sunlight, the light from the sun. "Will it stay this long?" Little asked. Big set a linen notebook beside the garden path. They had a question to keep.



Little took a piece of chalk, a soft stick for making marks. "First mark the tip," said Big. Little drew a small line where the shadow ended. To measure is to find how long something is. They used a measuring stick from the wooden stick's base to the chalk.



At noon, the middle of the day, they came back. The wooden stick had stayed in place, but its shadow had changed. "It is short and crisp!" said Little. Around noon, the sun is near its highest point for the day. The morning shadow had reached much farther.



"Which way does it point?" asked Big. Little studied the ground. Sunlight reached one side of the stick. The shadow stretched away on the other side. The bright side faced the sun; the shadow pointed away from the light. Little traced its edge along the ground with one finger.



Make a little book.

1. Print one-sided: US Letter, portrait, actual size / 100%. No fit. Headers/footers off.
2. Fold long edge to long edge; reopen. Fold short edge to short edge, then in half again. Open: eight boxes.
3. Cut only the solid center line across the middle two boxes. Keep the outer boxes joined.
4. Refold long edge to long edge, print outside. Push ends together to open a diamond, then a cross.
5. Wrap into a book with page 1 in front. Follow pages 1 to 8.



Measure, notice, and pause whenever you like.

Your turn

Write one command sentence about measuring a shadow.

Example: Measure from the stick's base to the shadow tip with the same unit.



In the notebook, Little drew a long line for morning and a short line for noon. Later, when the sun was lower, the shadow would grow longer again. A short noon shadow was a clue to a high sun. "What will we notice tomorrow?" Little asked.



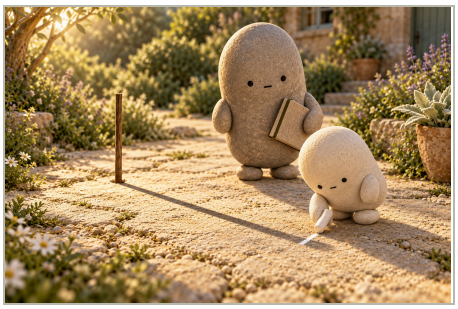
Little marked the new tip with chalk. They used the same measuring stick to check both lengths. They began at the same standing stick and used the same unit each time. The noon length was shorter. "The stick stayed the same," Little said. "The sun's height changed."

Give a measurement its words

Shadows at Noon · Curiosity day

Name _____ Date _____

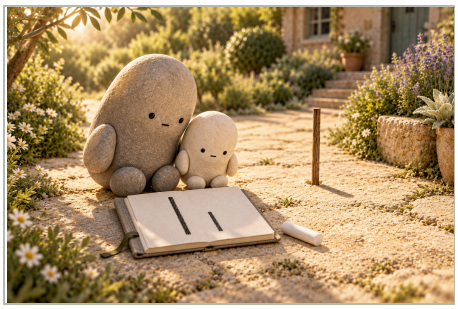
Write one clear command sentence about measuring a shadow. Say it first if you like.



Save the tip



Compare the marks



A clue to keep

Draw the stick, its shadow, and a chalk mark. Add one label.

stand · stick · mark · tip · shadow · measure · first · then

First, ... tell how to stand the stick or mark the tip

Then, ... tell how to measure from the stick to the mark

At the end, ... tell how to save or compare the length

Does your sentence tell what to do, with a command such as Stand, Mark, Measure, or Compare?